HANDS-ON LAB 02 GUIDE & COPY-PASTE PROMPTS • TELECOMMUNICATIONS EDITION (TELECOM NEXUS)

Delivered by Tryolabs • Powered by Google Antigravity & Gemini Enterprise

TRYOLABS • GOOGLE ANTIGRAVITY WORKSHOP • LAB 02 (EN)

# Lab 02: Encoding Telecom Domain Standards with Rules & Custom Skills

**Duration**: 45 minutes

**Delivered by**: **Tryolabs** (Powered by Google Antigravity & Gemini Enterprise)

**Antigravity Capabilities Practiced**: Workspace Rules (`GEMINI.md`, `.agents/rules/*.md`), Progressive-Disclosure Skills (`.agents/skills/<name>/SKILL.md`), `@` Mentions, `/plan`, `/learn`, and `/grill-me`.

■ **Quick Prompts File**: See PROMPTS.md (Español) for copy-paste prompts.

## 1. Scenario & Architectural Goal

In Lab 01, you fixed two costly OSS/BSS bugs manually with Antigravity. How do you ensure that **every developer and every AI agent session** across your Telecom engineering organization automatically enforces:

- **3GPP TS 28.552** (5G RAN PRB `< 80%` and PDCP latency SLA rules)
- **3GPP TS 32.298 & GSMA TAP3.12** (`Decimal(1024 ** 3)` GiB volume conversion & `NR_5G` visiting PLMN roaming markup)
- **TMF622** (Product Ordering status lifecycle)
- **CPNI Subscriber Privacy** (Masking IMSI and MSISDN in diagnostic outputs)

In **Google Antigravity**, you achieve this using two complementary customization primitives:

1. **Rules (`GEMINI.md` and `.agents/rules/telecom-standards.md`)**: Hierarchical guardrails loaded automatically to enforce coding styles, domain math (`Decimal`), and 3GPP/TMF compliance.
2. **Skills (`.agents/skills/sla-revenue-audit/SKILL.md`)**: On-demand runbooks using **Progressive Disclosure** (only the YAML `name` and `description` are loaded into context initially; the full runbook and helper script `scripts/run_sla_audit.py` are loaded only when the user or agent invokes the skill).

## 2. Lab Setup

Run the Lab 02 installer (which automatically ensures Lab 01 solutions are applied first):

```
./labs/lab-02-rules-and-skills/install.sh
```

## 3. Step-by-Step Instructions

### Step 3.1: Author `GEMINI.md` at the Repository Root

Use Antigravity (see PROMPTS.md) to create `GEMINI.md` at the root of the project. Ensure it specifies:

- The OSS (`ran_telemetry`, `ran_alarms.jsonl`) and BSS (`b2b_contracts`, `cdr_records`, `tmf622_sales_orders`) architecture.
- Mandatory Python `Decimal` (`ROUND_HALF_UP` to `0.01`) for all currency calculations.
- Binary GiB conversion (`1024 ** 3`) and `3GPP` slice types (`s_nssai_sst` 1, 2, 3).

### Step 3.2: Create `.agents/rules/telecom-standards.md`

Create `.agents/rules/telecom-standards.md` documenting:

- `3GPP` TS 28.552 and TS 32.298 standards
- `TAP3` roaming markup rules across `NR_5G` and `EUTRAN_4G`
- `TMF622` order feasibility classifications (`READY_TO_ACTIVATE`, `REQUIRES_SLICE_REMEDIATION`, `REJECTED`)
- `Decimal` financial precision and CPNI IMSI/MSISDN privacy safeguards

### Step 3.3: Create `.agents/skills/sla-revenue-audit/SKILL.md`

Create `.agents/skills/sla-revenue-audit/SKILL.md` starting with YAML frontmatter:

```
---
name: sla-revenue-audit
description: >-
  Executes a combined 5G Network Slice SLA compliance and GSMA TAP3 CDR revenue assuranc \
  e
  audit across OSS and BSS, generating an executive risk matrix in JSON and Markdown for \
  mats.
---
```

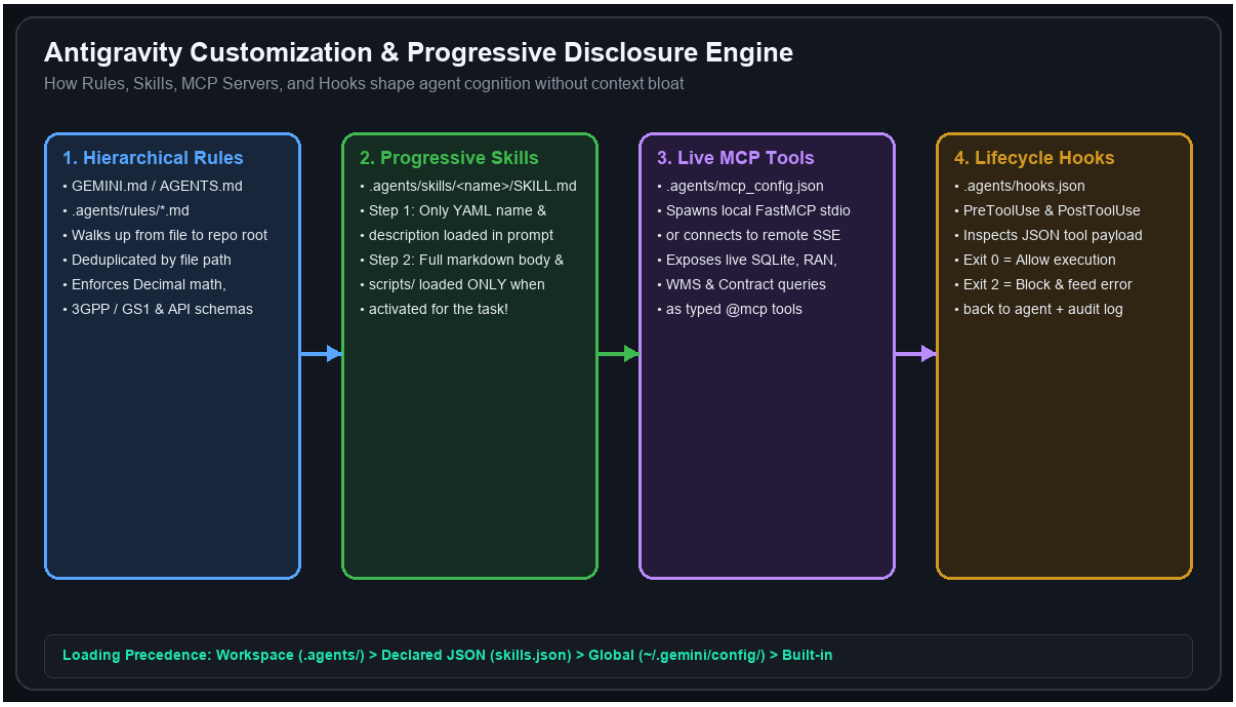
In the body of `SKILL.md`, instruct the agent to run `./.venv/bin/python scripts/run_sla_audit.py` and inspect `data/exports/sla_revenue_audit_report.json` and `data/exports/sla_revenue_audit_report.md`.

## 4. Verification

Run the Lab 02 verification script:

```
./labs/lab-02-rules-and-skills/verify.sh
```

**Need the Reference Solution?** Run `./labs/lab-02-rules-and-skills/apply_solution.sh`.


Antigravity Customization & Progressive Disclosure Engine
How Rules, Skills, MCP Servers, and Hooks shape agent cognition without context bloat
1. Hierarchical Rules
• GEMINI.md / AGENTS.md
• .agents/rules/*.md
• Walks up from file to repo root
• Deduplicated by file path
• Enforces Decimal math,
• 3GPP / GS1 & API schemas
2. Progressive Skills
• .agents/skills/<name>/SKILL.md
• Step 1: Only YAML name &
• description loaded in prompt
• Step 2: Full markdown body &
• scripts/ loaded ONLY when
• activated for the task!
3. Live MCP Tools
• .agents/mcp_config.json
• Spawns local FastMCP stdio
• or connects to remote SSE
• Exposes live SQLite, RAN,
• WMS & Contract queries
• as typed @mcp tools
4. Lifecycle Hooks
• .agents/hooks.json
• PreToolUse & PostToolUse
• Inspects JSON tool payload
• Exit 0 = Allow execution
• Exit 2 = Block & feed error
• back to agent + audit log
Loading Precedence: Workspace (.agents/) > Declared JSON (skills.json) > Global (~/.gemini/config/) > Built-in

**LAB 02 COPY-PASTE ANTIGRAVITY PROMPTS (EN)**
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# Lab 02 — Optimized Antigravity Prompts Cheat Sheet (English)

**Goal**: Use **Google Antigravity** (associated with your **Gemini Enterprise** license and delivered by **Tryolabs**) to encode Telecom 3GPP/GSMA/TMF domain governance into:

1. Root Workspace Constitution (`GEMINI.md`)
2. Modular Rule (`.agents/rules/telecom-standards.md`)
3. Progressive-Disclosure On-Demand Skill (`.agents/skills/sla-revenue-audit/SKILL.md`)

## 1. Discovery & Architecture Survey Prompt (`@` Mentions)

```
@src/services/contract_auditor.py @scripts/run_sla_audit.py @tests/test_lab02_rules_skil \
  ls.py
Inspect how Telecom Nexus combines 3GPP 5G RAN telemetry (`ran_telemetry`), GSMA TAP3 CD \
  R revenue assurance (`cdr_records`), and Enterprise B2B SLA contracts (`b2b_contracts` \
  ) to generate executive risk reports. Also inspect what `tests/test_lab02_rules_skills \
  .py` requires for `GEMINI.md`, `.agents/rules/telecom-standards.md`, and `.agents/skil \
  ls/sla-revenue-audit/SKILL.md`.
```

## 2. Planning Prompt (`/plan`)

```
/plan Design the three Antigravity Customization files required for Lab 02:
1. `GEMINI.md` at the repository root defining the OSS/BSS domain architecture, mandator \
  y Python `Decimal` arithmetic (`ROUND_HALF_UP`), `1024 ** 3` GiB scaling, and `ACTIVE` \
   contract SLA filtering.
2. `.agents/rules/telecom-standards.md` enforcing 3GPP TS 28.552 (5G RAN PRB < 80%, PDCP \
   latency), 3GPP TS 32.298 / GSMA TAP3.12, TMF622 order statuses, `Decimal` billing pre \
  cision, and CPNI IMSI/MSISDN privacy protection.
3. `.agents/skills/sla-revenue-audit/SKILL.md` with valid YAML frontmatter (`name: sla-r \
  evenue-audit`, `description: ...`) instructing Antigravity how to execute `scripts/run \
  _sla_audit.py` and validate `data/exports/sla_revenue_audit_report.json` and `.md`.
```

## 3. Instructive Inline Edit Prompt (`Cmd+I` / `Ctrl+I`)

Open `.agents/skills/sla-revenue-audit/SKILL.md` in Antigravity IDE and press `Cmd+I` / `Ctrl+I`:

```
Generate a complete Antigravity SKILL.md with YAML frontmatter (`name: sla-revenue-audit \
  ` and a third-person `description` explaining when to activate the skill) followed by  \
  a 4-step runbook that runs `./.venv/bin/pytest tests/test_lab01_triage.py`, executes ` \
  ./.venv/bin/python scripts/run_sla_audit.py`, and verifies `data/exports/sla_revenue_a \
  udit_report.json` and `data/exports/sla_revenue_audit_report.md`.
```

## 4. Implementation & Skill Invocation Prompt

First, ask Antigravity to create the three files:

```
Create `GEMINI.md`, `.agents/rules/telecom-standards.md`, and `.agents/skills/sla-revenu \
  e-audit/SKILL.md` according to our plan, then run `./labs/lab-02-rules-and-skills/veri \
  fy.sh`.
```

Next, test invoking your newly created **Skill** in natural language:

```
Use the `sla-revenue-audit` skill to run a full 5G Slice SLA & GSMA TAP3 CDR revenue ass \
  urance audit and summarize the top 3 highest-risk B2B enterprise contracts from the ge \
  nerated executive report.
```

# 5. Power-User & Stretch Prompts (`/grill-me`, `/learn`, `/compact`)

## Governance & Rule Design Interview (`/grill-me`)

```
/grill-me Interview me to refine our `.agents/rules/telecom-standards.md` so it covers 3 \
  GPP Release 18 Network Data Analytics Function (NWDAF), CPNI subscriber privacy maskin \
  g, and TMF622/TMF641 service order decomposition.
```

## Persist Rule Corrections (`/learn`)

```
/learn Whenever generating executive B2B SLA reports in Telecom Nexus, always sort contr \
  acts by `total_commercial_risk_usd` descending and highlight both `degraded_cell_id` a \
  nd `primary_tac`.
```